Consider Oxygen
Titrate to maintain SpO2 >94%
Routine Oxygen administration should be avoided

EMR

Check Glucose
<70mg/dL or >250mg/dL

Go to Hypo/ Hyperglycemia CPG

Avoid treating hypertension unless:
BP >220/120mmHg **with** signs of end organ dysfunction

Consider Tele EMS

Perform BFAST assessment

B – Balance (sudden loss of balance)
E – Eye (vision loss)
F – Face (facial drop or uneven smile)
A – Arm (arm weakness)
S – Speech (slurred speech)
T – Time (last time seen normal)

Consider Stroke Mimics

- Hypoglycemia
- Intracerebral lesions (e.g. tumor, abscess)
- Seizures and post-ictal states (Todd's paralysis)
- Hemiplegic migraine
- Electrolyte abnormalities (e.g. Hyponatremia)
- Conversion disorder (acute psycho-social stressors)

Obtain 12-Lead ECG
But do not delay transport

EMT

IV Access
Minimum 18g IV is preferred if possible

EMT

Position patient at 15°-30° head up (semi-reclined) unless the patient is required to be supine for airway management

Hospital Destination Decision
(as directed by dispatch 2)
Onset of Symptoms / last known well <4.5 hours

< 4.5 hours
Stroke Center

> 4.5 hours
Other Appropriate Hospital

Patient Care Goals

1. Rapidly identify suspected acute stroke or transient ischemic attack (TIA) to facilitate time-sensitive transfer.
2. Minimize secondary brain injury through supportive care.
3. Ensure rapid transport to a stroke-capable medical facility for definitive evaluation and treatment within the therapeutic window.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients presenting with signs and symptoms suggestive of acute stroke or TIA, including:
  - Sudden onset of focal neurological deficits (e.g., weakness, numbness, or paralysis of face, arm, or leg).
  - Speech difficulties (e.g., slurred speech, aphasia, or inability to speak).
  - Vision changes (e.g., sudden loss of vision, double vision).
  - Dizziness, loss of balance, or coordination problems.
  - Severe headache with no known cause (e.g., suggestive of hemorrhagic stroke).
- Symptoms consistent with stroke assessment tools (e.g., BEFAST).

Exclusion Criteria

- Symptoms clearly attributable to another cause (e.g., hypoglycemia, seizure, or trauma) without stroke-like features.
- Patients with resolved symptoms and no ongoing neurological deficits (e.g., prior TIA with no new symptoms).

Key Patient Assessment Points

1. **Stroke Screening**
   - Perform a 'BEFAST" Stroke Assessment:
     - **Balance:** Sudden loss of balance
     - **Eyes:** Vision Loss
     - **Face:** Check for facial drooping or asymmetry.
     - **Arms:** Assess for unilateral weakness or drift.
     - **Speech**: Evaluate for slurred speech or inability to speak clearly.
     - **Time**: Determine the exact time of symptom onset or last known well.
2. **Neurological Deficit Assessment**
   - Identify focal neurological signs (e.g., unilateral weakness, sensory loss, or neglect).
   - Assess for gaze deviation, pupil asymmetry, or facial paralysis.
   - Use AVPU (Alert, Verbal, Painful, Unresponsive) or Glasgow Coma Scale (GCS) to quantify level of consciousness.
3. **Time of Onset**
   - Confirm the precise time symptoms began or when the patient was last known well (critical for thrombolytic or endovascular therapy eligibility).
4. **Glucose Assessment**
   - Check random blood sugar (RBS) to rule out hypoglycemia (<70 mg/dL) or hyperglycemia (>400 mg/dL), which can mimic stroke symptoms.
5. **Stroke Mimics**
   - Consider Stoke Mimics:
     - Intracerebral lesions (e.g. tumor, abscess)
     - Seizures and post-ictal states (Todd's paralysis)
     - Hemiplegic migraine
     - Electrolyte abnormalities (e.g. Hyponatremia)
     - Conversion disorder (acute psycho-social stressors)
6. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.
     - All patients with signs of acute stroke are Red Priority.

Adult Patient Management & Interventions

1. **Primary & Secondary Survey**
   - Perform Primary and Secondary Surveys as per Universal Care CPG.
2. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
3. **Patient Positioning**
   - Patients should be positioned semi-recumbent or with the head elevated 15-30° if alert.
4. **Consider Oxygen**
   - Administer to maintain SpO2 ≥94% if hypoxia is present.
   - Avoid routine oxygen administration in normoxic patients.
5. **Consider Red Flags**
   - Onset of symptoms within 4.5 hours of arrival (thrombolytic window) or up to 24 hours (endovascular therapy window).
   - Rapidly deteriorating neurological status (e.g., worsening GCS, loss of consciousness).
   - Signs of increased intracranial pressure (e.g., unequal pupils, severe headache, vomiting).
   - Respiratory distress or hypoxia (SpO2 <90%).
   - Suspected hemorrhagic stroke (e.g., sudden severe headache, seizures).
6. **12-Lead ECG**
   - Perform a 12-lead ECG. However, do not delay transfer to acquire a 12-lead ECG if acute stroke is suspected, perform the 12-Lead en route to hospital.
7. **Establish IV Access**
   - Indicated for patients requiring fluid administration or at risk of deterioration.
   - A large bore IV cannula (18g or larger) in a large peripheral vein is preferred, when possible. An18g cannula (or larger) is required for administration of contrast CT scans.
   - Do not delay time on scene to establish IV access.
8. **Patient Reassessment**
   - Frequently Reassess the patient as per Universal Care CPG.

Patient Safety Considerations

- **Time-Sensitive Transport**
  - Patients with Stroke symptoms with an onset <4.5 hours should be rapidly transported to a stroke center.
  - Clearly communicate with Dispatch 2 identifying stroke symptoms and symptom onset time.
- **Time of Onset**
  - It is imperative that the last known well (LKW) time is accurately identified — the moment the patient was last known normal— as this dictates eligibility for interventions and will determine the correct hospital for the patient.

Clinical Insights & Additional Guidance

- **Blood Pressure Management**
  - Blood pressure should not be lowered unless it exceeds 220/120 mmHg with signs of end organ dysfunction. This preserves cerebral perfusion in ischemic stroke, where autoregulation is impaired.
  - Tele-EMS should be contacted for further advice on blood pressure management of exceptionally hypertensive patients.

Documentation

- Identification: Patient's full name, age, and gender.
- Mechanism / Medical Complaint: Sudden onset of neurological deficits, BEFAST symptoms, or witnessed stroke-like event.
- Injuries / Information: Stroke mimics considered and ruled out (e.g., glucose level, seizure history), presence of risk factors (e.g., hypertension, diabetes).

- Signs: Neurological exam findings (e.g., facial droop, limb weakness, slurred speech), GCS, pupil findings, vital signs, and RBS results.
- Treatment:
    - Time of symptom onset or last known well.
    - Oxygen administration and SpO2 levels.
    - IV access placement and cannula size.
    - Any interventions performed (e.g., positioning, blood pressure support).
    - Stroke alert activation and hospital notification.
- Allergies: Known drug or contrast allergies.
- Medications: Document chronic medications (e.g., anticoagulants, antiplatelets).
- Background: Relevant medical history (e.g., previous stroke, cardiovascular disease, atrial fibrillation).
- Other:
    - Scene findings and witnesses' statements.
    - Accurate documentation of symptom timeline.
    - Communication with Dispatch and receiving hospital.

Key Performance Indicators

1. Perform and document a BEFAST assessment for all suspected stroke cases.
2. Perform an RBS for all patients with suspected stroke.
3. Initiate transport all patients with stroke symptoms <4.5 hours from onset are transported within 15 minutes of EMS arrival.

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